



AMERICAN UNIVERSITY OF PHNOM PENH

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COSC 251 Spring 2026: Group Presentation and Coding Project (100 points)

Deadline: **Sep 23, 11:59 PM**

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Overview: For this group project, you will study and research a **sorting algorithm**, implement it in **Java**, compare it with basic sorting algorithms, write a short research report, create presentation slides, and record a video presentation. Your group should understand how the algorithm works, its characteristics, its time complexity, its performance on different dataset sizes, and when it should or should not be used.

This project is worth **10% of your total course grade**.

Project Requirements:

1. Each group must select **one sorting algorithm** and inform the instructor of your choice in the Telegram group by **Sep 14**. *No two groups may select the same algorithm.*
2. Each group must prepare a **20–30 minute video presentation** on their chosen algorithm.
3. The video presentation must be submitted to **Canvas** and also shared in the course **Telegram group** by **Sep 23, 11:59 PM**.
4. Each group must **code the selected sorting algorithm in Java**. The implementation must be written by the group and must not simply call a built-in sorting function such as `Arrays.sort()` or `Collections.sort()`.
5. The submitted Java code must include:
 - A Java implementation of the selected sorting algorithm.
 - Java implementations of **Bubble Sort**, **Selection Sort**, and **Insertion Sort** for comparison.
 - A test program or **main** method that allows the instructor to run and test the algorithms.
 - Test cases for the required dataset sizes:
 - **Small dataset:** 5 elements
 - **Medium dataset:** 1,000 elements
 - **Large dataset:** 1,000,000 elements
 - Output saving the sorted results to a txt file.
 - Performance results for each dataset size, such as runtime, number of comparisons, number of swaps or movements, or other relevant metrics.
 - Clear instructions explaining how to compile and run the program.
6. The program should be organized clearly so that the instructor can easily test the selected algorithm and compare it with Bubble Sort, Selection Sort, and Insertion Sort using the same input datasets.
7. For each dataset size, the program should show performance results such as:
 - Runtime
 - Number of comparisons

- Number of swaps or movements
 - Any other meaningful metric related to the selected algorithm
8. Each group must compare the selected sorting algorithm with the following basic sorting algorithms:
- Bubble Sort
 - Selection Sort
 - Insertion Sort

The comparison should explain how the selected algorithm performs compared to the basic algorithms, especially in terms of runtime, number of operations, time complexity, and suitability for small, medium, and large datasets.

9. Your research report, presentation slides, and video presentation must include the following:
- (a) A clear explanation of the key characteristics of the sorting algorithm.
 - (b) A step-by-step demonstration of how the algorithm works on a small dataset.
 - (c) An analysis of the algorithm's time complexity, including **best-case**, **average-case**, and **worst-case**.
 - (d) A discussion of situations where the algorithm performs well and why.
 - (e) A discussion of situations where the algorithm should **not** be used and why.
 - (f) Real-world examples where the algorithm is applied and reasons for choosing it over other algorithms.
 - (g) A description of the Java implementation.
 - (h) Testing results for the small, medium, and large datasets.
 - (i) Runtime, comparisons, swaps, movements, or other relevant performance metrics.
 - (j) A comparison with Bubble Sort, Selection Sort, and Insertion Sort.
10. The **research report** should be a well-structured document that includes:
- Introduction
 - Algorithm explanation with examples
 - Time complexity analysis
 - Java implementation
 - Testing methodology
 - Performance results for 5, 1,000, and 1,000,000 elements
 - Comparison with Bubble Sort, Selection Sort, and Insertion Sort
 - Appropriate and inappropriate use cases
 - Real-world applications
 - Conclusion and findings

Diagrams, pseudocode, tables, screenshots, charts, or illustrations should be included where appropriate.

11. The **presentation slides** should summarize your research clearly and concisely. Slides should be well-organized, visually supportive, and easy to follow. The slides must include the algorithm explanation, time complexity, implementation summary, testing results, and comparison with Bubble Sort, Selection Sort, and Insertion Sort.
12. The **video presentation** should clearly explain the selected sorting algorithm using the presentation slides. The video should also include a demonstration of the Java code and the testing results. The presentation should be understandable, well-structured, and should show that all group members understand the topic.

13. All group members must demonstrate understanding of the algorithm. *You are not required to memorize code, but you must understand how the algorithm works, how it was implemented, and how to interpret the testing results.*

What to Submit:

This is a **group assignment**. Each group should submit **one submission** on Canvas by **June 23, 11:59 PM**.

1. Presentation slides
2. Research report in PDF format
3. Video presentation
4. Java source code, including:
 - The selected sorting algorithm
 - Bubble Sort
 - Selection Sort
 - Insertion Sort
 - Test code for small, medium, and large datasets
5. Instructions explaining how to compile and run the Java program

In addition, each group must also share the **video presentation** in the course **Telegram group** by **June 23, 11:59 PM**.

Grading Criteria:

1. **Correctness and Understanding (20 points):** Accuracy and understanding of the algorithm, time complexity, and performance behavior.
2. **Requirement Satisfaction (30 points):** All required research, report, presentation, implementation, testing, and comparison topics are addressed.
3. **Java Implementation and Testing (25 points):** The selected sorting algorithm and the required basic sorting algorithms are correctly implemented, tested on the required dataset sizes, and submitted with clear running instructions.
4. **Organization and Report Quality (10 points):** Clear structure, logical flow, well-written report, and effective use of diagrams, tables, or examples.
5. **Presentation Delivery (15 points):** Clear, confident, and understandable video presentation with good explanation of the algorithm, implementation, and findings.